



Retifanlimab with carboplatin and paclitaxel for locally recurrent or metastatic squamous cell carcinoma of the anal canal (POD1UM-303/InterAACT-2): a global, phase 3 randomised controlled trial



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Summary

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Background Retifanlimab has activity in programmed death ligand 1-positive advanced squamous cell anal carcinoma (SCAC) that has progressed on platinum chemotherapy. We aimed to prospectively assess the benefit of adding retifanlimab to initial carboplatin–paclitaxel for this disease.

Methods This global, multicentre, double-blind, randomised, controlled, phase 3 trial was done at 70 centres in 12 countries across the EU, Australia, Japan, the UK, and the USA. Patients aged ≥18 years with inoperable locally recurrent or metastatic SCAC, an Eastern Cooperative Oncology Group performance status of 0 or 1, no previous systemic therapy, and well controlled HIV (ie, CD4+ count >200/μL and undetectable viral load) were eligible. Patients were randomly assigned (1:1) to retifanlimab (500 mg intravenous) or placebo every 4 weeks with standard carboplatin–paclitaxel for up to 1 year. Patients in the placebo group could cross over to retifanlimab monotherapy on confirmed disease progression. The primary endpoint was independently assessed progression-free survival (ie, time from date of randomisation to date of first documented progressive disease or death due to any cause) per Response Evaluation Criteria in Solid Tumours version 1.1. Efficacy was assessed by intention to treat. This trial is registered with ClinicalTrials.gov (NCT04472429) and EUDRA-CT (2020–000826–24) and is active but closed to enrolment.

Findings Between Nov 12, 2020, and July 3, 2023, 376 patients were assessed for eligibility and 308 were randomly assigned to retifanlimab plus carboplatin–paclitaxel (n=154) or placebo plus carboplatin–paclitaxel (n=154). 222 (72%) of 308 patients were female and 86 (28%) were male. Median progression-free survival was 9·3 months (95% CI 7·5–11·3) in the retifanlimab group and 7·4 months (7·1–7·7) in the placebo group (hazard ratio 0·63 [95% CI 0·47–0·84]; one-sided p=0·0006). Serious and grade 3 or worse adverse events were more frequent in the retifanlimab plus carboplatin–paclitaxel group compared with the placebo plus carboplatin–paclitaxel group (47·4% vs 38·8% and 83·1% vs 75·0%, respectively). The most common grade ≥3 adverse events were neutropenia (35·1% for retifanlimab plus carboplatin–paclitaxel vs 29·6% for placebo plus carboplatin–paclitaxel) and anaemia (19·5% vs 20·4%). Four fatal adverse events occurred in the retifanlimab plus carboplatin–paclitaxel group, only one (pancytopenia) of which was treatment related. One fatal adverse event occurred in the placebo plus carboplatin–paclitaxel group and was not treatment related.

Interpretation Retifanlimab provides clinical benefit, with a manageable safety profile, when added to first-line chemotherapy in advanced squamous cell carcinoma of the anal canal. These results suggest retifanlimab with carboplatin plus paclitaxel should be considered as the new standard of care for patients with advanced squamous cell anal carcinoma.

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Introduction

Squamous cell carcinoma of the anal canal (SCAC) accounts for 2·5% of all gastrointestinal malignancies.¹ The incidence of anal cancer has increased by approximately 3% per year over the past decade, mainly driven by endemic human papillomavirus (HPV)

infection.^{2,3} Immunosuppression, particularly with concurrent HIV infection, is an important amplifier of risk due to inhibition of primary HPV infection clearance.⁴ Individuals with HIV are 25-fold to 35-fold more likely to develop SCAC than those who are HIV negative.⁵

Research in context

Evidence before this study

We searched ClinicalTrials.gov from database inception to Nov 12, 2024, for phase 3 clinical trials of systemic therapies for the treatment of advanced squamous cell carcinoma of the anal canal (SCAC) using the keywords “squamous cell carcinoma of the anal canal” or “anal cancer, squamous cell carcinoma” (Condition/disease) and “systemic therapy” (Other terms), and filtered by phase 3 studies. To our knowledge, there are no US Food and Drug Administration-approved therapies, and treatment options for patients with advanced SCAC have remained largely unchanged over several decades due to few studies being done in this patient population. Carboplatin with weekly paclitaxel is the current recommended regimen based on the phase 2 InterAACT trial results and European Society for Medical Oncology and National Comprehensive Cancer Network guidelines. However, the progression-free survival and overall survival with this regimen are short (8·1 months and 20·0 months, respectively). Retifanlimab is an anti-programmed cell death-1 (PD-1) humanised monoclonal antibody that is approved in the USA and Europe for advanced Merkel cell carcinoma. Retifanlimab showed improved clinical responses and a favourable safety profile compared with previous salvage chemotherapy in a phase 2 study of patients with platinum-refractory advanced SCAC. The aforementioned studies have led to the phase 3 study of retifanlimab in combination with standard-of-care chemotherapy in patients with advanced SCAC reported here.

Added value of this study

The phase 3 POD1UM-303/InterAACT-2 study evaluated the ability of retifanlimab to provide additive benefit to standard-of-care chemotherapy in patients with locally advanced or

metastatic SCAC not previously treated with systemic therapy.

Although the current standard-of-care platinum-based chemotherapy provides high initial response rates, patients commonly experience disease progression and have short median survival times, highlighting an unmet medical need. To our knowledge, this is the first and largest well controlled trial in recurrent or metastatic SCAC, a rare disease.

The addition of retifanlimab to carboplatin–paclitaxel led to significantly longer median progression-free survival than placebo plus carboplatin–paclitaxel, with a consistent benefit in favour of retifanlimab plus chemotherapy observed for all subgroup comparisons. At the interim analysis, despite the immature data, results suggested a possible improvement of overall survival. Similar rates of carboplatin–paclitaxel dose interruption were observed in both groups, with higher rates of serious and grade ≥ 3 adverse events observed with retifanlimab that could be managed with standard measures and did not compromise administration of carboplatin–paclitaxel.

Implications of all the available evidence

Data from our study demonstrate clinically meaningful improvements in survival and tumour control with retifanlimab plus carboplatin–paclitaxel compared with the current preferred carboplatin plus paclitaxel regimen in patients with previously untreated advanced SCAC. Retifanlimab with carboplatin plus paclitaxel should be considered as the new standard of care for this difficult-to-treat disease. Future studies could investigate the combination of retifanlimab with novel targeted agents added to standard-of-care chemotherapy for the treatment of patients with advanced SCAC including those with HIV who are most at risk for SCAC.

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See Online for appendix

Approximately 14% of anal cancers are metastatic at diagnosis, and approximately 40% of patients with localised SCAC will progress after initial chemoradiation.^{6,7} The 5-year relative survival rate is approximately 36% for patients with distant metastasis.^{6–8} In addition to compromised survival, patients with advanced anal cancer report bleeding, rectal and abdominal pain, and feeling stigmatised, all of which lead to a diminished quality of life.^{9,10} To date, no phase 3 trials of systemic therapy have been conducted in recurrent or metastatic SCAC, and there are no US Food and Drug Administration-approved therapies, but the European Society for Medical Oncology guidelines and National Comprehensive Cancer Network guidelines recommend carboplatin with weekly paclitaxel as the preferred standard of care based on the phase 2 InterAACT trial results.^{11,12} However, median progression-free survival and overall survival with this regimen are still modest, at 8·1 months and 20·0 months, respectively.¹³

HPV-driven malignancy is an active area of research and a promising target for immunotherapy based on the

biology of HPV.^{14–17} The combination of a programmed cell death-1 (PD-1) inhibitor with platinum-based chemotherapy has demonstrated improved survival in other HPV-driven cancers, specifically cervical cancer and squamous cancer of the head and neck.^{18–20} Retifanlimab is an anti-PD-1 humanised monoclonal antibody,^{21–25} which is approved in the USA and Europe for advanced Merkel cell carcinoma.²⁶ In a phase 2 study of patients with platinum-refractory advanced SCAC, retifanlimab showed improved clinical responses and a favourable safety profile compared with previous salvage chemotherapy alone.²⁷ Responses were observed regardless of programmed death ligand 1 (PD-L1) expression, liver metastases, or HIV status.²⁷ Based on these data, retifanlimab has been recommended as a treatment option for platinum-refractory advanced SCAC.¹¹ Preliminary data in patients with refractory cervical cancer and penile cancer (another rare HPV-driven malignancy) show anti-tumour activity with retifanlimab, supporting further study of this novel PD-1 inhibitor in SCAC.^{28,29}

This phase 3 trial aimed to prospectively evaluate whether retifanlimab can provide additive benefit to standard-of-care chemotherapy in patients who have

locally advanced or metastatic SCAC not previously treated with systemic therapy.

Methods

Study design

POD1UM-303/InterAACT-2 is a global, multicentre, double-blind, randomised, controlled, phase 3 trial done at 70 centres in 12 countries across the EU, Australia, Japan, the UK, and the USA. The trial protocol including all amendments (appendix p 15) was approved by local institutional review boards (appendix p 5). The study was conducted in accordance with the Declaration of Helsinki, Good Clinical Practice, and applicable local laws and specific regulations. Ethics approval was obtained through the National Research Ethics Service Committee London. Study design and conduct were guided by an external steering committee; safety was monitored by an external data monitoring committee. Various stakeholders (eg, public, professionals, and regulatory bodies) provided input in the design of the study protocol. This study is registered with ClinicalTrials.gov (NCT04472429) and EUDRA-CT (2020-000826-24) and is open but closed to enrolment.

Patients

Patients aged ≥ 18 years with inoperable locally recurrent or metastatic SCAC, an Eastern Cooperative Oncology Group performance status of 0 or 1, no previous systemic therapy, and well controlled HIV (ie, CD4+ count $>200/\mu\text{L}$ and undetectable viral load) were eligible. Previous chemotherapy administered concomitantly with radiotherapy or previous neoadjuvant or adjuvant therapy if completed 6 months or more before study entry was permitted. Patients with previous PD-1-directed and PD-L1-directed therapy were excluded. Patients were recruited via the following methods, including but not limited to academic institution oncology databases, letters to community oncologists and health-care practitioners, HIV SCAC advocacy referrals, mini awareness calls, and during large annual conferences.

Demographic data including sex were collected from medical records and if not available, site staff asked patients to self-declare, and it was documented in an ongoing medical chart. Some demographic data, such as ethnicity and race, were scarce due to local and/or country privacy laws. Options provided in the data capture system were male and female. All patients provided written informed consent. A full list of eligibility criteria is provided in the study protocol.

Randomisation and masking

Patients were randomly assigned (1:1) to retifanlimab or placebo plus the standard regimen of carboplatin and paclitaxel¹³ using an interactive web response system to conceal allocation. Randomisation was stratified by PD-L1 expression using the SP263 PD-L1 (Ventana) assay (Roche Diagnostics, IN, USA; $<1\%$ or $\geq 1\%$), disease status (locally

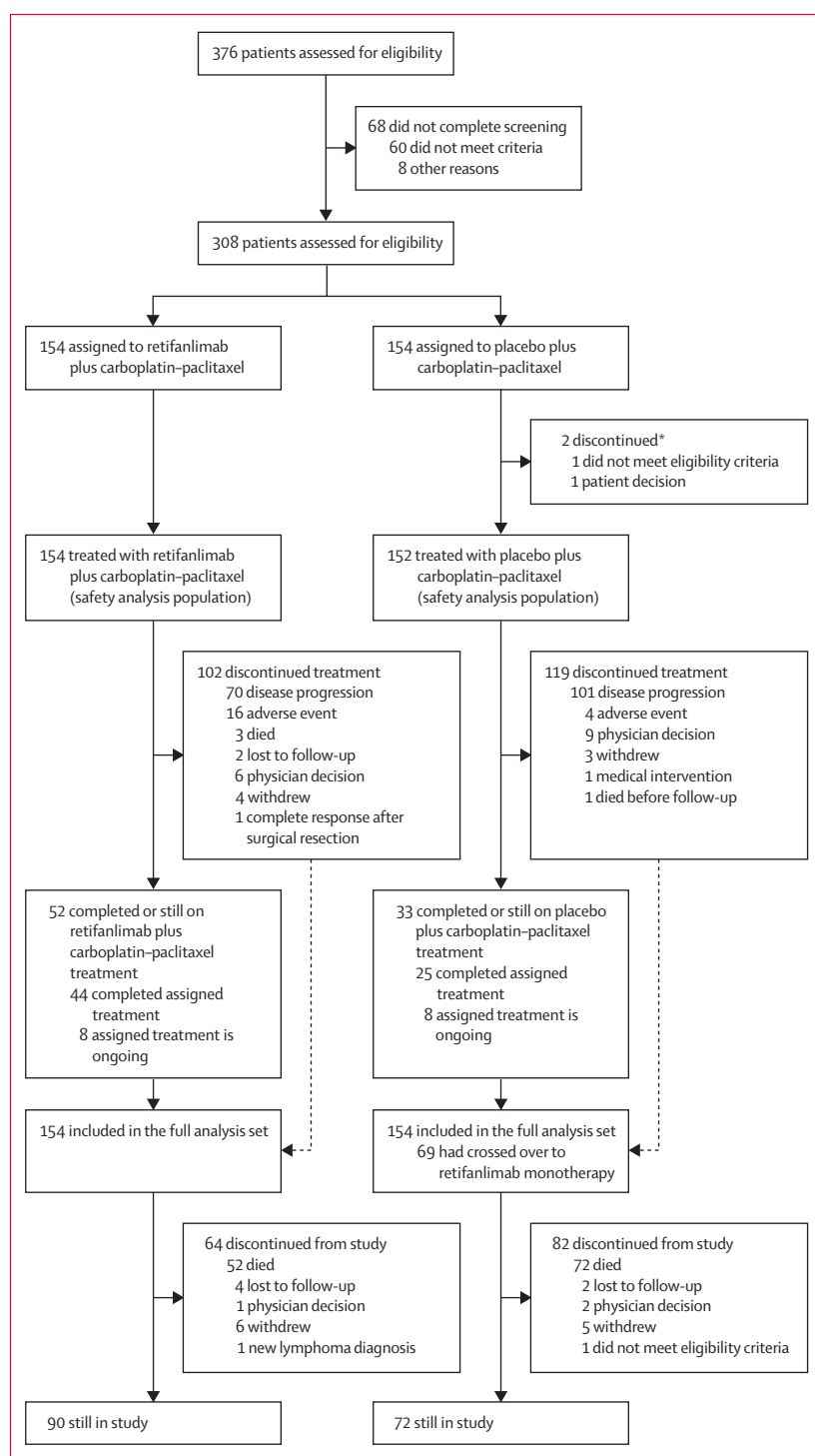


Figure 1: Trial profile

*Two patients included in full analysis set for overall survival.

recurrent or metastatic), and region (Australia, the EU, North America, the UK, or rest of the world). Patients in the placebo plus carboplatin–paclitaxel group with disease progression verified by a masked central radiology team could cross over to retifanlimab monotherapy. Patients, investigators, and the sponsor remained masked to patient assignment throughout the randomised phase of the study.

Procedures

Patients received up to six cycles (24 weeks) of carboplatin (area under the curve 5 mg/mL on day 1) and paclitaxel (80 mg/m² on days 1, 8, and 15), both administered as intravenous infusions. Retifanlimab 500 mg or placebo was administered as a 30-min intravenous infusion before scheduled carboplatin–paclitaxel on day 1 of each 28-day cycle, or as monotherapy (crossover period) for up to 1 year, until disease progression, unacceptable toxicity, withdrawal of consent, or loss to follow-up. The schedule of assessments and visits is shown in the study protocol.

Outcomes

The primary endpoint was progression-free survival, defined as the time from the date of randomisation to the date of the first documented disease progression or death due to any cause. Progression-free survival was assessed by masked independent central review using Response Evaluation Criteria in Solid Tumours version 1.1, with imaging performed every 8 weeks. Secondary endpoints were overall survival (key secondary endpoint), overall response rate by masked independent central review, duration of response, disease control rate, pharmacokinetics, and safety. Safety was evaluated by the number of patients having adverse events and the number of patients discontinuing retifanlimab or placebo due to adverse events. Pharmacokinetic data and final analysis of overall survival will be reported in a subsequent publication.

Statistical analysis

The study was designed to have >80% power to demonstrate a clinically meaningful improvement of progression-free survival (hazard ratio [HR] 0·67) at a one-sided overall 2·5% level of significance, with a planned enrolment of approximately 300 patients (predicted 207 events). Efficacy was evaluated in the full analysis set, which included all randomly assigned patients, according to treatment group and stratification at randomisation. Safety was evaluated in the safety population, which included all randomly assigned patients who received at least one dose of study drug.

Survival data were analysed using the Kaplan–Meier method. Progression-free and overall survival were further analysed using a stratified log-rank test at an overall one-sided 2·5% level of significance following a gatekeeping procedure to ensure type 1 error control. HRs and 95% CIs were estimated using stratified Cox regression. One-sided p values <0·025 were considered statistically significant. The difference of overall response

rate was calculated using the Cochran–Mantel–Haenszel test. A Kaplan–Meier estimate of the distribution function was constructed for duration of response. The rank-preserving structure failure time model and inverse property of censoring weighting methods were used to adjust for a crossover effect on overall survival. Crossover was part of the study design and was optional; it was funded as part of the trial design and procedures. Strategies to reduce dropout and censoring included study newsletters describing the study endpoints, designing the procedures for patient convenience, and reviewing clinical data in real time. An external data monitoring committee monitored safety and efficacy throughout the study. Statistical analyses were performed using SAS, version 9.4.

	Retifanlimab plus carboplatin– paclitaxel (n=154)	Placebo plus carboplatin– paclitaxel (n=154)
Age, years	62 (29–86)	61 (35–84)
Sex		
Female	104 (68%)	118 (77%)
Male	50 (32%)	36 (23%)
Race		
White	132 (86%)	137 (89%)
Black or African American	3 (2%)	2 (1%)
Asian	10 (6%)	8 (5%)
Other	3 (2%)	2 (1%)
Geographical location		
Australia, the EU, North America, or the UK	146 (95%)	146 (95%)
Rest of the world	8 (5%)	8 (5%)
Previous radiotherapy	104 (68%)	113 (73%)
Locally recurrent disease	28 (18%)	27 (18%)
Metastatic disease*	126 (82%)	127 (82%)
Liver	55 (36%)	56 (36%)
ECOG PS 0	82 (53%)	86 (56%)
HIV positive	6 (4%)	5 (3%)
HPV status†		
Positive	29 (19%)	28 (18%)
Negative	3 (2%)	4 (3%)
Unknown or missing	122 (79%)	122 (79%)
P16 status‡		
Positive	76 (49%)	74 (48%)
Negative	16 (10%)	15 (10%)
Indeterminate or missing	62 (40%)	65 (42%)
PD-L1 expression status ≥1*§	139 (91%)	140 (91%)

Data are median (range) or n (%). ECOG PS=Eastern Cooperative Oncology Group performance status. HPV=human papillomavirus. PD-L1=programmed cell death ligand 1. *Stratification factor. †Samples available from local testing.

‡Samples available from central testing. §PD-L1 expression <1 also includes patients with non-evaluable tumour tissue; PD-L1 expression data were unavailable for three patients in the retifanlimab plus carboplatin–paclitaxel group and seven in the placebo plus carboplatin–paclitaxel group.

Table 1: Baseline demographics and disease characteristics

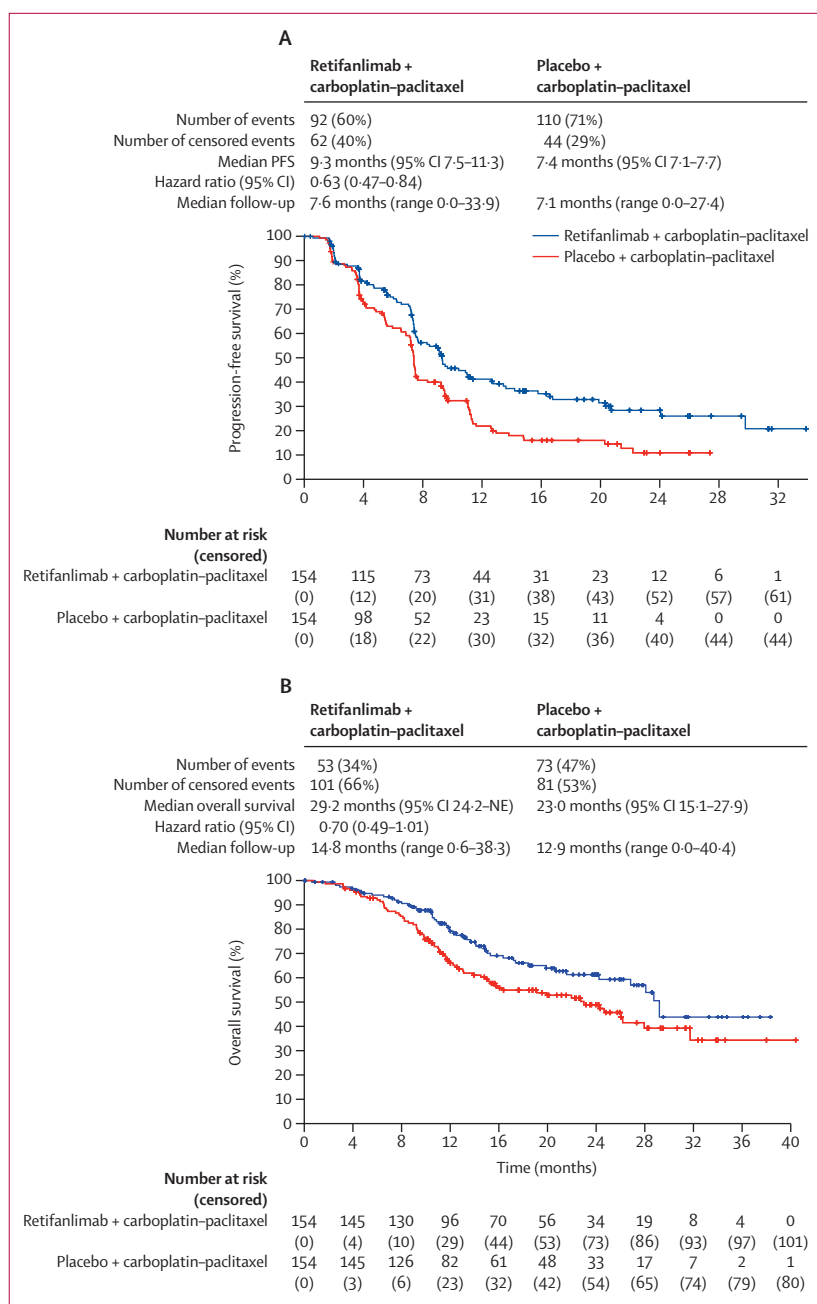


Figure 2: Kaplan-Meier analysis of progression-free survival (A) and overall survival (B) in the full analysis set
NE=not evaluable. PFS=progression-free survival.

Role of the funding source

The sponsor of the study had a role in the study design, data collection, data analysis, and writing of the report.

Results

Between Nov 12, 2020, and July 3, 2023, 376 patients were assessed for eligibility and 308 were randomly assigned to retifanlimab plus carboplatin-paclitaxel (n=154) or placebo plus carboplatin-paclitaxel (n=154). At data cutoff for the primary analysis (April 15, 2024),

306 patients had been treated with retifanlimab plus carboplatin-paclitaxel (n=154) or placebo plus carboplatin-paclitaxel (n=152; figure 1). In the crossover period, 69 (45%) of 154 patients randomly assigned to placebo plus carboplatin-paclitaxel crossed over to retifanlimab monotherapy. Baseline demographics and disease characteristics were balanced between the treatment groups (table 1). 222 (72%) of 308 patients were female and 86 (28%) were male. Six (4%) of 154 patients in the retifanlimab plus carboplatin-paclitaxel group and five (3%) in the placebo plus carboplatin-paclitaxel group were HIV positive. Most tumour samples had PD-L1 expression $\geq 1\%$ (139 [90%] of 154 patients in the retifanlimab plus carboplatin-paclitaxel group and 140 [91%] of 154 patients in the placebo plus carboplatin-paclitaxel group).

202 progression-free survival events had been recorded at data cutoff. The trial met its primary endpoint. Median progression-free survival was significantly longer in the retifanlimab plus carboplatin-paclitaxel group than in the placebo plus carboplatin-paclitaxel group (9.3 months [95% CI 7.5–11.3] vs 7.4 months [95% CI 7.1–7.7]; HR 0.63 [95% CI 0.47–0.84]; $p=0.0006$), with a median follow-up of 7.6 months (range 0–33.9) and 7.1 months (0–27.4), respectively (figure 2A). A consistent benefit in favour of retifanlimab plus carboplatin-paclitaxel was observed for all predefined subgroups with sufficient patients for comparison (figure 3).

At time of progression-free survival analysis, median overall survival was 29.2 months (95% CI 24.2 to not evaluable) in the retifanlimab plus carboplatin-paclitaxel group and 23.0 months (15.1 to 27.9) in the placebo plus carboplatin-paclitaxel group (figure 2B). There was some evidence of improved survival despite survival data being relatively immature and the high proportion of placebo plus carboplatin-paclitaxel patients who entered the crossover period (HR 0.70 [95% CI 0.49–1.01]; $p=0.027$). Additionally, across all predefined subgroups, a consistent benefit in overall survival in favour of retifanlimab plus carboplatin-paclitaxel was observed, except for patients who had PD-L1 negative tumours and those who were HIV positive (appendix p 12). Sensitivity analyses adjusting for the crossover effect demonstrated further robustness of the overall survival benefit for retifanlimab plus carboplatin-paclitaxel and indicated that crossover did not have a meaningful effect on the results (appendix p 13). Patients continue to be followed for subsequent final overall survival analysis.

An overall response rate of 55.8% (95% CI 47.6–63.8) was observed in the retifanlimab plus carboplatin-paclitaxel group versus 44.2% (36.2–52.4) with placebo plus carboplatin-paclitaxel ($p_{\text{nominal}}=0.013$). 34 (22%) of 154 patients in the retifanlimab plus carboplatin-paclitaxel group had a complete response, 52 (34%) had a partial response, and 45 (29%) had stable disease, with a disease control rate of 87.0% (95% CI 80.7–91.9).

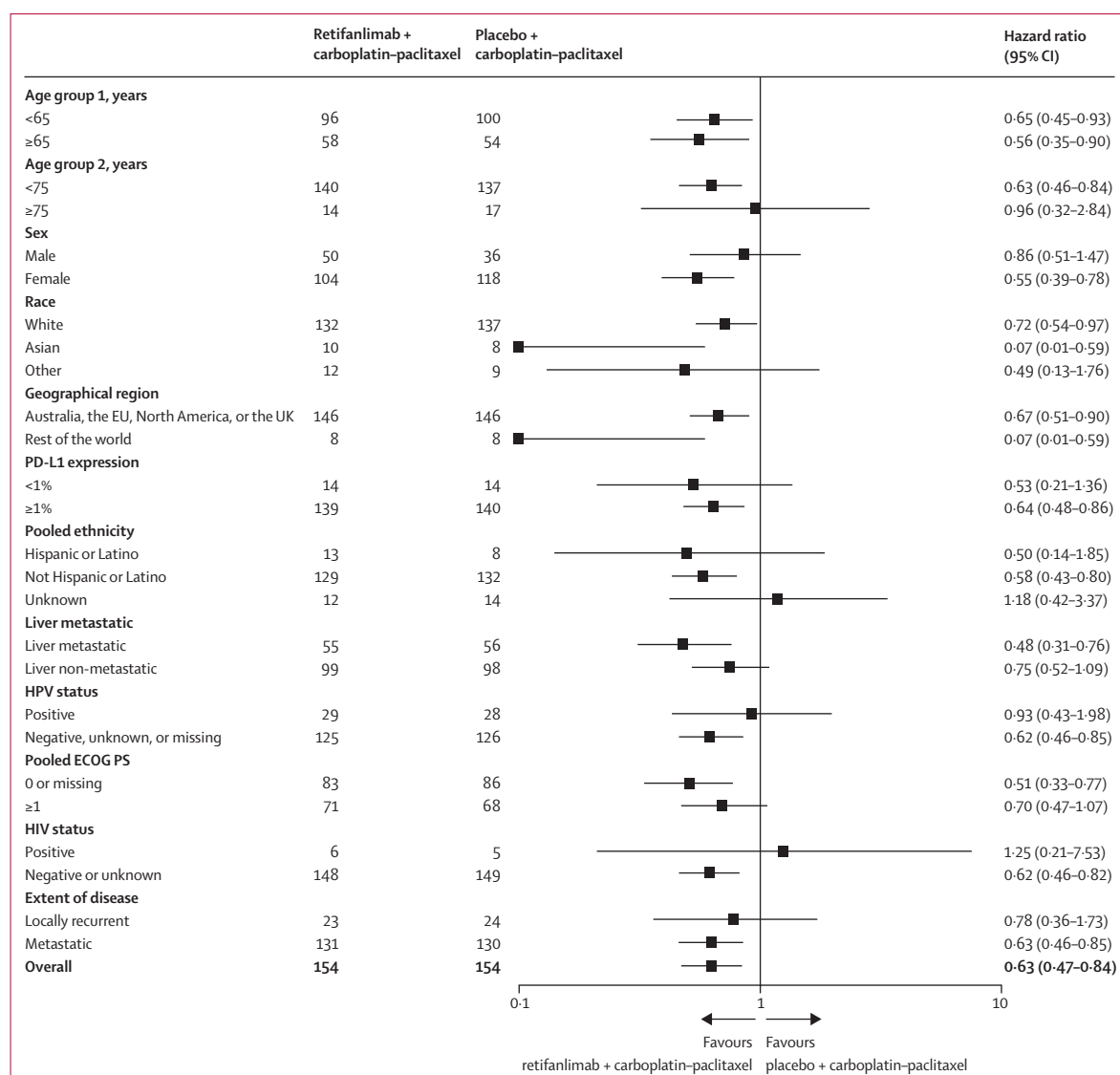


Figure 3: Forest plot of progression-free survival by masked independent central review in the full analysis set

ECOG PS=Eastern Cooperative Oncology Group performance status. HPV=human papillomavirus. PD-L1=programmed death ligand 1.

21 (14%) of 154 patients in the placebo plus carboplatin-paclitaxel group had a complete response, 47 (31%) had a partial response, and 52 (34%) had stable disease, with a disease control rate of 79.9% (95% CI 72.7–85.9). Median duration of response was 14.0 months (95% CI 8.6–22.2) in the retifanlimab plus carboplatin-paclitaxel group and 7.2 months (5.6–9.3) in the placebo plus carboplatin-paclitaxel group (appendix p 14).

Carboplatin-paclitaxel administration was not compromised by the addition of retifanlimab to the regimen. Patients in both treatment groups received a median of six (range 1–6) carboplatin infusions and 15 (1–18) paclitaxel infusions. Patients received a median of nine (1–13) retifanlimab infusions in the retifanlimab plus carboplatin-paclitaxel group and eight (1–13) placebo

infusions in the placebo plus carboplatin-paclitaxel group. Similar rates of carboplatin-paclitaxel dose interruption were observed in both groups (112 [73%] of 154 patients in the retifanlimab plus carboplatin-paclitaxel group and 111 [73%] of 152 in the placebo plus carboplatin-paclitaxel group). The rate of carboplatin-paclitaxel discontinuation was higher in the retifanlimab plus carboplatin-paclitaxel group (16 [10%] patients) than in the placebo plus carboplatin-paclitaxel group (eight [5%] patients), mostly due to the expected increase in recognised immune-related toxicities (eg, endocrinopathy and colitis).

Serious adverse events and grade 3 or worse adverse events were more frequent in the retifanlimab plus carboplatin-paclitaxel group than in the placebo plus carboplatin-paclitaxel group (table 2).

	Retifanlimab plus carboplatin– paclitaxel (n=154)	Placebo plus carboplatin– paclitaxel (n=152)	Total (n=306)
Treatment duration, months	7·4 (0·03–14·6)	6·8 (0·03–14·6)	7·2 (0·03–14·6)
Patients with any adverse events	154 (100%)	152 (100%)	306 (100%)
Patients with grade ≥ 3 adverse events	128 (83%)	114 (75%)	242 (79%)
Deaths	4 (3%)*	1 (1%)†	5 (2%)
Patients with serious adverse events	73 (47%)	59 (39%)	132 (43%)
Treatment-related serious adverse events	25 (16%)	10 (7%)	35 (11%)
Immune-related adverse events	75 (49%)	40 (26%)	115 (38%)
Adverse events leading to discontinuation of retifanlimab or placebo	17 (11%)	4 (3%)	21 (7%)

Data are median (range) or n (%). *Events were metastases to peritoneum (n=1), pancytopenia (n=1), pneumonia (n=1), and sepsis (n=1). †Fatal pneumonia.

Table 2: Summary of safety

Retifanlimab-related serious and grade 3 or worse adverse events were observed in 25 (16%) of 154 patients and 52 (34%) patients in the retifanlimab plus carboplatin–paclitaxel group compared with ten (7%) of 152 patients and 39 (26%) patients in the placebo plus carboplatin–paclitaxel group, respectively. Four fatal adverse events were observed in the retifanlimab plus carboplatin–paclitaxel group, one of which (pancytopenia) was attributed to retifanlimab by the investigator. This patient had decompensated hepatic cirrhosis at study entry, which probably contributed to their poor tolerance of treatment. One fatal adverse event occurred in the placebo plus carboplatin–paclitaxel group, but this was not treatment related. Immune-related adverse events occurred in 75 (49%) of 154 patients in the retifanlimab plus carboplatin–paclitaxel group and 40 (26%) of 152 in the placebo plus carboplatin–paclitaxel group; most of these events were grade 1 or 2 in severity and did not lead to dose interruption, delay, or discontinuation. Most common adverse events and immune-related events are summarised in the appendix (pp 9–10). No loss of HIV control or opportunistic infections among patients known to be HIV positive was observed.

Among patients who crossed over to retifanlimab monotherapy, no deaths, serious retifanlimab-related events, or adverse events leading to retifanlimab infusion interruption occurred (appendix p 11). Immune-related events were reported in 14 (20%) of 69 patients; most were grade 1 or 2 in severity. Infusion-related reactions occurred in two (3%) of 69 patients, and all were grades 1 or 2.

Discussion

This global, multicentre, double-blind, randomised, phase 3 trial evaluated the efficacy and safety of retifanlimab in combination with standard carboplatin–paclitaxel in patients with inoperable locally advanced or metastatic SCAC who had not received previous systemic chemotherapy. To our knowledge, this is the first and

largest prospectively designed phase 3 trial of systemic therapy to be completed in patients with advanced SCAC. Although considered a rare malignancy, the incidence and prevalence of SCAC are increasing worldwide due to its association with endemic HPV infection.^{2,3} Despite this, treatment options for SCAC have remained largely unchanged over several decades.³⁰ Even though the current standard of care (platinum-based chemotherapy) provides high initial response rates, subsequent disease progression is common and median survival times are short,¹³ highlighting the unmet need for improved treatment options for patients with advanced disease.

Retifanlimab plus carboplatin–paclitaxel demonstrated clinically meaningful improvement in progression-free survival, overall survival, and tumour control compared with the current carboplatin plus paclitaxel standard of care. Importantly, Kaplan–Meier curves for progression-free survival and overall survival appear to separate early and remain apart for the entire treatment interval, suggesting that most patients were likely to have benefited from retifanlimab. Although cross-study comparisons have limitations, the patient population enrolled in this POD1UM-303/InterAACT-2 trial was similar to the phase 2 InterAACT trial, as were the outcomes in the carboplatin–paclitaxel control group,¹³ providing additional validation for our study results. The progression-free survival outcomes were also generalisable to predefined subgroups of interest, with the possible exception of patients who were HIV positive, for whom the 95% CIs around the point estimate are extremely wide. Given the previously reported benefits of retifanlimab for patients who were HIV positive in the POD1UM-202 study,²⁷ the lack of clear benefit in patients who were HIV positive in our study might reflect the small number of events recorded. However, enrolment of this population should be considered an imperative for future SCAC trials. Patients with liver metastases (who are historically resistant to immunotherapy) also appear to have benefited, as has previously been reported with retifanlimab monotherapy in platinum-refractory SCAC.²⁷

Our results contrast with a recent phase 2 study that evaluated the PD-L1 inhibitor atezolizumab plus a different chemotherapy regimen (modified docetaxel, cisplatin, and fluorouracil regimen) in chemotherapy-naïve patients with metastatic or unresectable locally advanced recurrent SCAC.³¹ This trial did not meet its primary endpoint of improved investigator-assessed progression-free survival at 12 months, and the toxicity of the combination was notable. Based on these results, the rationale for triplet chemotherapy as a backbone for further development has been questioned.³² An ongoing trial of nivolumab with the carboplatin–paclitaxel doublet (NCT04444921) will provide additional information on the less intensive approach we have selected.

Retifanlimab plus carboplatin–paclitaxel was generally well tolerated in the present study. Despite the higher rate of serious, grade 3 or worse, and fatal adverse events

with retifanlimab plus carboplatin–paclitaxel than placebo plus carboplatin–paclitaxel, these toxicities were manageable with standard measures, and carboplatin–paclitaxel delivery was not compromised. Our results are consistent with the extensive data for other combinations of platinum-based chemotherapy with PD-1 and PD-L1 inhibitors.³³ Moreover, safety data in the crossover period were generally similar to previous reports for PD-1 and PD-L1 monotherapy.²⁶ Notably, treatment was also well tolerated in patients who were HIV positive, who are the group most at risk for SCAC.⁵

The principal limitation of this study is the small sample size, which reflects the rarity of the disease but limits interpretation of some of the subgroup data. We also did not collect detailed information on the initial presentation of SCAC (eg, perianal tumours). We are not aware, however, of any previous trials that approach the scope of POD1UM-303/InterAACT-2, so our results provide an important benchmark against which future studies in this neglected disease might be conducted.

In conclusion, retifanlimab improves clinical outcomes with a manageable safety profile when added to standard-of-care chemotherapy in advanced SCAC. Retifanlimab with carboplatin plus paclitaxel should be considered the new standard of care for this difficult-to-treat disease.

Contributors

SR was involved in study design, study conduct, patient recruitment, and data collection. ES-S, LE, MBA, FM, AR, LD, ST, ASD, MPS, NC, RG, AL, JM, MF, J-PS, and PR were responsible for the recruitment of patients, trial conduct, and data collection. All authors contributed to the interpretation of the data, drafting and critical review of the manuscript, and provided approval of the final version to be published. All authors had full access to all the data in the study and final responsibility for the decision to submit for publication. SR, CT, JH, MMJ, and MC had direct access to and verified the underlying data reported in the manuscript.

Declaration of interests

SR reports an advisory role within AstraZeneca, Bayer, BeiGene, Hookipa, Incyte, Merck Serono, Nordic Pharma, Seagen, Servier, and Takeda; expert testimony to Boehringer Ingelheim; speaker invitations from Bayer, Merck Serono, and Servier; and travel grants from Incyte, Servier and BMS. ES-S reports an advisory role within Astellas, BeiGene, BMS, and Pierre Fabre Oncology; speakers bureau participation for BMS, MSD, Pierre Fabre Oncology, and Servier; and receiving travel grants from Amgen, Merck Serono, MSD, and Pierre Fabre Oncology. LE reports an advisory role within Astellas; and honoraria from AstraZeneca, Servier, BMS, MSD, Merck Serono, and Pierre Fabre. MBA reports consultancy work for Merck Serono; membership of an advisory board within Amgen, AstraZeneca, Bayer, BMS, Incyte, Merck, Pierre Fabre, and Servier; and a travel grant from Roche. FM reports honoraria from Pierre Fabre and Servier; travel grants from Amgen, Pierre Fabre, and Servier; and an institutional research grant from Incyte. AR reports an advisory role with AstraZeneca, MSD, BMS, BeiGene, Greywolf Therapeutics, and Medison Pharma; expert testimony for AstraZeneca; speaker invitations from Pierre Fabre, MSD, BMS, Ipsen, Servier, and Limbic Oncology; and institutional research grants from Specialised Therapeutics and Merck. LD reports honoraria from Amgen, BMS, Mylan, OSÉUS Médical, and Servier. ASD reports a travel grant from Novartis. MPS reports honoraria from Merck, Servier, GSK, Incyte, Nordic, Takeda, and Bayer. NC reports a travel grant from GenesisCare. RG reports honoraria from Advanced Accelerator Applications, Amgen, AstraZeneca, Bayer, BMS, Esteve, MSD, Novartis, Pierre Fabre, Roche, Servier, and Viatriis; an advisory role with BMS (Mexico), MSD, and Pierre Fabre; and travel grants from Advanced Accelerator Applications, Amgen, MSD Oncology, Mylan,

Pierre Fabre, and Merck. AL reports honoraria from Amgen, Astellas, AstraZeneca, BMS (Mexico), Esteve, Ipsen, LEO Pharma, MSD, Mylan, Pierre Fabre, Servier, and Viatriis; an advisory role with Astellas Pharma, Bayer, BMS (Mexico), Incyte, Pierre Fabre, Servier, and Takeda; institutional research funding from Bayer and Lilly; and travel grants from Bayer, Esteve, Ipsen, Pierre Fabre, Servier, and Viatriis. JM reports an advisory role with Advance Medical, Cancer Expert Now, Delcath Systems, and MSD; research grants from Amgen, Guardant Health, Incyte, Merck, and Terumo; travel grants from Amgen; and a leadership role with Grupo Español Multidisciplinar en Cáncer Digestivo. MF reports consultancy work with AbbVie, Adagene, AstraZeneca, Bayer, BMS, Incyte, Merck, Mirati, Pfizer, and Taiho; and an advisory role with Eisai, Entos, Janssen, Nouscom, Roche/Genentech, and Tempus. CT, JH, MMJ, and MC report employment with and stock ownership in Incyte. J-PS reports consultancy work with Incyte and Merck-MSD; an advisory board role within AstraZeneca, Daiichi Sankyo, Exact-Science, Gilead, Lilly, Novartis, Pierre Fabre Oncology, Pfizer, and Roche; and research grants from Incyte, LEO Pharma, and MSD Avenir. PR reports an advisory board role within Servier and travel grants from Amgen, Sanofi, and Servier. ST declares no competing interests.

Data sharing

Incyte (Wilmington, DE, USA) is committed to data sharing that advances science and medicine while protecting patient privacy. Qualified external scientific researchers may request anonymised datasets owned by Incyte for the purpose of conducting legitimate scientific research. Researchers may request anonymised datasets from any interventional study (except phase 1 studies) for which the product and indication have been approved on or after Jan 1, 2020, in at least one major market (eg, the USA, Europe, or Japan). Data will be available for request after the primary publication or 2 years after the study has ended. Information on Incyte's clinical trial data sharing policy and instructions for submitting clinical trial data requests are available at: <https://www.incyte.com/Portals/0/Assets/Compliance%20and%20Transparency/clinical-trial-data-sharing.pdf?ver=2020-05-21-132838-960>.

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